United States Patent Application Publication	20250283599
Kind Code	A1
Publication Date	September 11, 2025
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DILUTION PASSAGES FOR COMBUSTOR OF A GAS TURBINE ENGINE

Abstract

A wall assembly for use in a combustor of a gas turbine engine, the wall assembly including: a support shell; a liner panel; and an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage; a backstop extending from the top portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

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Family ID: **94173245**

Appl. No.: **18/596302**

Filed: **March 05, 2024**

Publication Classification

Int. Cl.: **F23R3/00** (20060101)

U.S. Cl.:

CPC **F23R3/002** (20130101); F23R2900/03042 (20130101)

Background/Summary

BACKGROUND

[0001] The present disclosure relates to gas turbine engines and, more particularly, to a combustor section therefor.

[0002] Gas turbine engines typically include a compressor section, a combustor section and a turbine section. During operation, air is pressurized in the compressor section and is mixed with fuel and burned in the combustor section to generate hot combustion gases. The hot combustion gases are communicated through the turbine section, which extracts energy from the hot combustion gases to power the compressor section and other gas turbine engine loads.

[0003] The combustor section typically includes an outer shell lined with heat shields often referred to as floatwall panels which are attached to the outer shell with studs and nuts. In certain arrangements, dilution holes in the floatwall panel communicate with respective dilution holes in the outer shell to direct cooling air for dilution of the combustion gases. In addition to the dilution holes, the outer shell may also have relatively smaller air impingement holes to direct cooling air between the floatwall panels and the outer shell to cool the cold side of the floatwall panels. This cooling air exits effusion holes on the surface of the floatwall panels to form a film on a hot side of the floatwall panels which serves as a barrier against thermal damage.

[0004] One particular region where localized hot spots may arise is around the combustor dilution holes. The dilution holes inject relative lower temperature air into the swirling fuel-rich cross flow for combustion. As the air penetrates into the fuel-rich cross-stream, heat release takes place along the reaction front creating high temperature regions around the dilution holes. A stagnation region along the upstream side of the dilution jets also forms a higher pressure environment such that cross flow momentum deflects the incoming dilution jet. It is the combination of high pressure and the deflection of the incoming jet which is believed to create a high temperature recirculation region along the inner surface of the dilution hole. These high temperatures recirculation regions may affect the life of the turbine of the gas turbine engine as well as the cooling design of the turbine.

[0005] A lower velocity region of flow along the perimeter of the dilution hole may be highly susceptible to inflow of hot combustion gas products. The inflow of these products can occur within a localized ingestion region and may result in a durability concern because a low temperature boundary condition is replaced by high temperature gases.

BRIEF DESCRIPTION

[0006] Disclosed is a wall assembly for use in a combustor of a gas turbine engine, the wall assembly including: a support shell; a liner panel; and an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage; a backstop extending from the top portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

[0007] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an internal plenum is located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum.

[0008] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, a plurality of openings are located in the surface of the backstop above the support shell, the plurality of openings being in fluid communication with the internal plenum.
[0009] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, a protrusion extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion.
[0010] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the at least one opening is a plurality of individual openings or a single elongated slot.
[0011] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the protrusion has a curvature that matches a curvature of the dilution passage.
[0012] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of individual openings or the single elongated slot are arranged in an arc.
[0013] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the backstop is arranged to be at least partially located at a trailing edge portion of the dilution passage.
[0014] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, a plurality of openings are located in the surface of the backstop, a surface of the annular peripheral wall defining the dilution passage and a bottom portion of the annular peripheral wall, the plurality of opening being in fluid communication with the cooling airflow paths and the cooling airflow paths are angled with respect to the surface of the backstop and the bottom portion of the annular peripheral wall.
[0015] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, some of the plurality of openings located in the bottom portion the annular peripheral wall are further from a trailing edge of the dilution passage than other ones of the plurality of openings located in the bottom portion of the annular peripheral wall, the some of the plurality of openings located in the bottom portion of the annular peripheral wall that are further from the trailing edge of the dilution passage correspond to some of the plurality of openings located in the surface of the backstop or the surface of the annular peripheral wall that are further from the bottom portion of the annular peripheral wall than others of the plurality of openings located in the surface of the backstop or the surface of the annular peripheral wall.
[0016] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an internal plenum is located in the annular peripheral wall, the cooling airflow paths extending through the internal plenum and at least one opening located in a surface of the annular peripheral wall defining the dilution passage, the at least one opening being in fluid communication with the internal plenum; a protrusion extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion.
[0017] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the at least one opening is located in the trailing edge portion of the protrusion is a plurality of individual openings or a single elongated slot.
[0018] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the protrusion has a curvature that matches a curvature of the dilution passage.
[0019] In addition to one or more of the features described above, or as an alternative to any of the

foregoing embodiments, the plurality of individual openings or the single elongated slot are arranged in an arc.

[0020] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an internal plenum is located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum; and a plurality of openings are located in the surface of the backstop above the support shell, a surface of the annular peripheral wall defining the dilution passage and a bottom portion of the annular peripheral wall, the plurality of openings being in fluid communication with the internal plenum.

[0021] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, an internal plenum is located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum; a protrusion extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion; a plurality of openings located in the surface of the backstop above the support shell, and a surface of the annular peripheral wall defining the dilution passage, the plurality of openings and the at least one opening located in a trailing edge portion of the protrusion being in fluid communication with the internal plenum.

[0022] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the protrusion completely encircles the dilution passage and a bottom portion of the protrusion has openings in fluid communication with the internal plenum.

[0023] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the internal plenum extends completely around an interior of the annular peripheral wall.

[0024] Also disclosed is a gas turbine engine, including: a compressor section; a turbine section; and a combustor section, the combustor section including a wall assembly for use in a combustor of the combustor section, the wall assembly including: a support shell; a liner panel; and an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage; a backstop extending from the top portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

[0025] Also disclosed is a method of guiding airflow into a combustor of a gas turbine engine, the method including: locating a portion of an annular peripheral wall in an opening of a support shell of the combustor, the annular peripheral wall extending from a liner panel secured to the support shell, the annular peripheral wall defining a dilution passage; and directing airflow into the dilution passage via a backstop extending from a top portion of the annular peripheral wall, the top portion defining a portion of a periphery of the dilution passage; and directing airflow into cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

[0027] FIG. **1** is a schematic, partial cross-sectional view of a gas turbine engine in accordance with this disclosure;

[0028] FIG. **2** is schematic cross-section of another example gas turbine engine;

[0029] FIG. **3** is an expanded longitudinal schematic sectional view of a combustor section that may be used with the example gas turbine engine architectures shown in FIGS. **1** and **2**;

[0030] FIG. **4** is an exploded view of a wall assembly with a dilution passage in accordance with the present disclosure;

[0031] FIG. **5** is a cross-sectional view of a dilution passage in accordance with an embodiment of the present disclosure;

[0032] FIG. **6** is a view along lines **6-6** of FIG. **5**;

[0033] FIG. **7** is a cross-sectional view of a dilution passage in accordance with an alternative embodiment of the present disclosure;

[0034] FIG. **8** is a cross-sectional view of a dilution passage in accordance with an alternative embodiment of the present disclosure;

[0035] FIG. **9** is a view along lines **9-9** of FIG. **8**;

[0036] FIG. **10** is a cross-sectional view of a dilution passage in accordance with another embodiment of the present disclosure;

[0037] FIG. **11** is a view along lines **11-11** of FIG. **10** in accordance with another embodiment of the present disclosure;

[0038] FIG. **11**A is a view along lines **11-11** of FIG. **10** in accordance with another embodiment of the present disclosure; and

[0039] FIG. **11**B is a view along lines **11-11** of FIG. **10** in accordance with another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0040] A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the FIGS.

[0041] FIG. **1** schematically illustrates a gas turbine engine **20**. The gas turbine engine **20** is disclosed herein as a two-spool turbo fan that generally incorporates a fan section **22**, a compressor section **24**, a combustor section **26** and a turbine section **28**. Referring to FIG. **2**, alternative engine architectures **200** might include additional sections **12**, **14**, and **16** in addition to the fan section **22′**, compressor section **24′**, combustor section **26′** and turbine section **28′** among other systems or features. Referring again to FIG. **1**, the fan section **22** drives air along a bypass flowpath while the compressor section **24** drives air along a core flowpath for compression and communication into the combustor section **26** then expansion through the turbine section **28**. Although depicted as a turbofan in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with turbofans as the teachings may be applied to other types of turbine engines such as a turbojets, turboshafts, and three-spool (plus fan) turbofans wherein an intermediate spool includes an intermediate pressure compressor (“IPC”) between a low pressure compressor (“LPC”) and a high pressure compressor (“HPC”), and an intermediate pressure turbine (“IPT”) between a high pressure turbine (“HPT”) and a low pressure turbine (“LPT”).

[0042] The engine **20** generally includes a low spool **30** and a high spool **32** mounted for rotation about an engine central longitudinal axis A relative to an engine static structure **36** via several bearing structures **38**. The low spool **30** generally includes an inner shaft **40** that interconnects a fan **42**, a low pressure compressor (“LPC”) **44** and a low pressure turbine (“LPT”) **46**. The inner shaft **40** may drive the fan **42** directly, or through a geared architecture **48** as illustrated in FIG. **1** to drive the fan **42** at a lower speed than the low spool **30**. An exemplary reduction transmission is an epicyclic transmission, namely a planetary or star gear system.

[0043] The high spool **32** includes an outer shaft **50** that interconnects a high pressure compressor (“HPC”) **52** and a high pressure turbine (“HPT”) **54**. A combustor **56** is arranged between the high pressure compressor **52** and the high pressure turbine **54**. The inner shaft **40** and the outer shaft **50** are concentric and rotate about the engine central longitudinal axis A which is collinear with their longitudinal axes.

[0044] Core airflow is compressed by the low pressure compressor **44** then the high pressure compressor **52**, mixed with the fuel and burned in the combustor **56**, then expanded over the high pressure turbine **54** and the low pressure turbine **46**. The low pressure turbine **46** and the high pressure turbine **54** rotationally drive the respective low spool **30** and the high spool **32** in response to the expansion. The main engine shafts **40**, **50** are supported at a plurality of points by the bearing structures **38** within the static structure **36**. It should be understood that various bearing structures **38** at various locations may alternatively or additionally be provided.

[0045] With reference to FIG. **3**, the combustor section **26**, **26**′ generally includes the combustor **56** with an outer combustor wall assembly **60**, an inner combustor wall assembly **62** and a diffuser case module **64** therearound. The outer combustor wall assembly **60** and the inner combustor wall assembly **62** are spaced apart such that an annular combustion chamber **66** is defined therebetween.

[0046] The outer combustor wall assembly **60** is spaced radially inward from an outer diffuser case **64**-O of the diffuser case module **64** to define an outer annular plenum **76**. The inner combustor wall assembly **62** is spaced radially outward from an inner diffuser case **64**-I of the diffuser case module **64** to define an inner annular plenum **78**. It should be understood that although a particular combustor is illustrated, other combustor types with various combustor liner arrangements will also benefit therefrom. It should be further understood that the disclosed cooling flow paths are but an illustrated embodiment and should not be limited only thereto.

[0047] The combustor wall assemblies **60**, **62** contain the combustion products for direction toward the turbine section **28**. Each combustor wall assembly **60**, **62** generally includes a respective support shell **68**, **70** which supports one or more liner panels **72**, **74** mounted thereto. Each of the liner panels **72**, **74** may be generally rectilinear and manufactured of, for example, a nickel based super alloy, ceramic or other temperature resistant material and are arranged to form a liner array. In the liner array, a multiple of forward liner panels **72**A and a multiple of aft liner panels **72**B are circumferentially staggered to line the outer shell **68**. A multiple of forward liner panels **74**A and a multiple of aft liner panels **74**B are circumferentially staggered to also line the inner shell **70**.

[0048] The combustor **56** further includes a forward assembly **80** immediately downstream of the compressor section **24** to receive compressed airflow therefrom. The forward assembly **80** generally includes an annular hood **82**, a bulkhead assembly **84**, and a multiple of swirlers **90** (one shown). Each of the swirlers **90** is circumferentially aligned with one of a multiple of fuel nozzles **86** (one shown) and the respective hood ports **94** to project through the bulkhead assembly **84**. The bulkhead assembly **84** includes a bulkhead support shell **96** secured to the combustor walls **60**, **62**, and a multiple of circumferentially distributed bulkhead liner panels **98** secured to the bulkhead support shell **96** around each respective swirler opening **92**. The bulkhead support shell **96** is generally annular and the multiple of circumferentially distributed bulkhead liner panels **98** are segmented, typically one to each fuel nozzle **86** and swirler **90**.

[0049] The annular hood **82** extends radially between, and is secured to, the forwardmost ends of the combustor wall assemblies **60**, **62**. The annular hood **82** includes the multiple of circumferentially distributed hood ports **94** that receive one of the respective multiple of fuel nozzles **86** and facilitates the direction of compressed air into the forward end of the combustion chamber **66** through a respective one of the swirler openings **92**. Each fuel nozzle **86** may be secured to the diffuser case module **64** and project through one of the hood ports **94** into the respective swirler **90**.

[0050] The forward assembly **80** introduces core combustion air into the forward section of the combustion chamber **66** while the remainder enters the outer annular plenum **76** and the inner annular plenum **78**. The multiple of fuel nozzles **86** and adjacent structure generate a blended fuel-air mixture that supports stable combustion in the combustion chamber **66**.
[0051] Opposite the forward assembly **80**, the outer and the inner support shells **68**, **70** are mounted adjacent to a first row of nozzle guide vanes (NGVs) **54**A in the high pressure turbine **54**. The nozzle guide vanes **54**A are static engine components which direct core airflow combustion gases onto the turbine blades of the first turbine rotor in the turbine section **28** to facilitate the conversion of pressure energy into kinetic energy. The core airflow combustion gases are also accelerated by the nozzle guide vanes **54**A because of their convergent shape and the gases are typically given a “spin” or a “swirl” in the direction of turbine rotor rotation. The turbine rotor blades absorb this energy to drive the turbine rotor at high speed.
[0052] With reference to FIG. **4**, a plurality of studs **100** extend from the liner panels **72**, **74** so as to permit the liner panels **72**, **74** to be mounted to their respective support shells **68**, **70** with fasteners **102** such as nuts. That is, the studs **100** project rigidly from the liner panels **72**, **74** and through the respective support shells **68**, **70** to receive the fasteners **102** at a threaded distal end section thereof.
[0053] A multiple of cooling impingement passages **104** penetrate through the support shells **68**, **70** to allow air from the respective annular plenums **76**, **78** to enter cavities **106**A, **106**B formed in the combustor wall assemblies **60**, **62** between the respective support shells **68**, **70** and liner panels **72**, **74**. The cooling impingement passages **104** are generally normal to the surface of the liner panels **72**, **74**. The air in the cavities **106**A, **106**B provides cold side impingement cooling of the liner panels **72**, **74**. As used herein, the term impingement cooling generally implies heat removal from a part via an impinging gas jet directed at a part.
[0054] A multiple of effusion passages **108** penetrate through each of the liner panels **72**, **74**. The geometry of the passages (e.g., diameter, shape, density, surface angle, incidence angle, etc.) as well as the location of the passages with respect to the high temperature main flow also contributes to effusion film cooling. The combination of impingement passages **104** and effusion passages **108** may be referred to as an Impingement Film Floatwall (IFF) assembly.
[0055] The effusion passages **108** allow the air to pass from the cavities **106**A, **106**B defined in part by a cold side **110** of the liner panels **72**, **74** to a hot side **112** of the liner panels **72**, **74** and thereby facilitate the formation of thin, cool, insulating blanket or film of cooling air along the hot side **112**. The effusion passages **108** are generally more numerous than the impingement passages **104** to promote the development of film cooling along the hot side **112** to sheath the liner panels **72**, **74**. Film cooling as defined herein is the introduction of a relatively cooler air at one or more discrete locations along a surface exposed to a high temperature environment to protect that surface in the region of the air injection as well as downstream thereof.
[0056] A plurality of dilution passages **116** penetrate through both the respective support shells **68**, **70** and liner panels **72**, **74** along a common axis D. For example only, in a Rich-Quench-Lean (R-Q-L) type combustor, the dilution passages **116** are located downstream of the forward assembly **80** to quench the hot combustion gases within the combustion chamber **66** by direct supply of cooling air from the respective annular plenums **76**, **78**. In one non-limiting embodiment, the dilution passages **116** are formed by an annular grommet or annular peripheral wall **115** extending upwardly from the cold side of the liner panels **72**, **74**. The annular peripheral wall **115** being received within an opening **117** in shells **68**, **70** when the liner panels **72**, **74** and the shells **68**,**70** are assembled together. The annular peripheral wall **115** terminates at a top portion **119** that defines a first periphery of the dilution opening **116** and a backstop or extended wall portion **121** extends upwardly from the top portion **119** of the annular peripheral wall **115**. In other words, top portion **119** forms a portion of a periphery of the dilution passage and the backstop **121** forms another portion or the remaining portion of a periphery of the dilution passage **116**. In one embodiment, the annular peripheral wall **115**, backstop **121** and liner panels **72**, **74** are integrally formed as a single

unitary structure or in other words, they are formed as a single piece.

[0057] With reference to FIG. **5** and as mentioned above, at least one of the multiple of dilution passages **116** includes a backstop **121** extending upwardly and away from the liner panels **72**, **74** proximate to a peripheral portion of the dilution passage **116**. The backstop **121** partially circumscribes each or a subset of the dilution passages **116**. As used herein a subset of dilution passages may also be referred to as a sector. In one non-limiting embodiment, the backstops **121** are symmetric. However, alternate embodiments could incorporate intentional asymmetries. For example, the backstop **121** could be rotated about the dilution passage centerline D. As illustrated in at least FIG. **4**, the backstop **121** may have a curvature that matches a curvature of the dilution opening **116**.

[0058] As illustrated and in one embodiment, the backstop **121** is arranged to be at least partially located at a trailing edge portion **125** of the dilution passage **116**. The trailing edge portion **125** of the dilution passage **116** is located opposite a leading edge portion **127** of the dilution passage **116**. As used herein, the “leading edge” refers to a portion of the dilution passage that faces the incoming airflow and the “trailing edge” refers to a portion of the dilution passage downstream from the “leading edge”. In other words, the leading edge encounters the incoming airflow prior to the trailing edge. In addition, and as used herein, “top” and “bottom” refer to heights or distances from or radially from the engine central longitudinal axis A. The top of an item is radially further from the engine central longitudinal axis A than a bottom of the same item.

[0059] As illustrated in FIG. **5**, the backstop **121** as well as the annular peripheral wall **115** has an interior plenum **130**. The interior plenum **130** is in fluid communication with a cooling airflow (illustrated by arrows **132**) via a plurality of openings **134**. The plurality of openings or at least a majority of the plurality of openings **134** are located in the backstop **121** above shells **68**, **70** so that they are positioned to capture cooling airflow flowing in the direction of arrows **132**, which may be located closer to a top **135** of the backstop **121**. This airflow above shells **68**, **70** and closer to a top **135** of the backstop **121** may be referred to as a velocity head of cooling airflow.

[0060] The annular peripheral wall **115** has a protrusion **136** that extends from a bottom portion **137** of the annular peripheral wall **115** past the hot side **112** of the liner panels **72**, **74**. The bottom portion **137** defines at least a portion of a second periphery of the dilution opening **116** and the protrusion **136** forms another portion or the remaining portion of a periphery of the second periphery of the dilution passage **116**.

[0061] The distance the protrusion **136** extends past the hot side **112** of the liner panels **72**, **74** is illustrated by arrows **138**. The protrusion **136** like the backstop **121** partially circumscribes each or a subset of the dilution passages **116**. As illustrated, the protrusion **136** like the backstop **121** may have a curvature that matches a curvature of the dilution opening **116**. The interior plenum **130** extends into the protrusion and is in fluid communication with the combustion chamber **66** via a plurality of or an array of individual openings **140** located in a trailing edge portion **142** of protrusion **136**. In an alternative embodiment, the plurality of individual openings **140** may be replaced by a single elongated slot or opening **140**.

[0062] As such, the internal plenum **130** via openings **134** and **140** provides cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall **115** to provide cooling air to a surface **112** of the liner panel **72**, **74**. The cooling air flow (illustrated by arrows **132**) will enter the interior plenum **130** from the backstop **121** and then exit the trailing edge portion **142** of the protrusion **136** via openings **140**. The exiting airflow is illustrated by arrows **144** This cooling air will provide surface film cooling to the hot side **112** of the liner panels **72**, **74**. This surface film cooling is illustrated by area **146**. This surface film cooling is downstream of the protrusion **136** and is particularly useful in cooling the hot side **112** of the liner panels **72**, **74** downstream of the protrusion **136**.

[0063] As illustrated in FIG. **6**, the openings or single elongated slot or opening **140** are arranged in an arc **147** in order to provide the surface film cooling illustrated by area **146**. This area **146** is located upstream of the surface cooling provided by openings **104** and **108**, which is illustrated by arrows **148**, **150**. As illustrated in FIG. **5**, the cooling openings **108** are angled in a downstream orientation to provide cooling air in the direction of arrows **150**. This area **146** helps to reduce hot spots adjacent to dilution passages **116** which are hard to cool as they are directly behind the also permits utilization of the relatively limited cooling air elsewhere in the combustor allowing for the more efficient use of combustor airflow. There is also dilution air or cooling air that is jetted into dilution passage **116** that does not enter internal plenum **130**. This air is illustrated by arrow **152**.

[0064] Referring now to FIG. **7**, an alternative embodiment of the present disclosure is illustrated. In this embodiment, there is no protrusion **136** or internal plenum **130**. However and in order to provide film cooling at the trailing edge **125** of opening **116**, an array or a plurality of cooling paths **154** extending through the backstop **121** as well as the annular peripheral wall **115** defining opening **116**. Each of these cooling paths **154** has an inlet opening **155** and an exit opening **157**. Each of the inlet openings **155** are arranged at various heights from the trailing edge **125** of a bottom of the opening **116**. Each of these inlet openings **155** correspond to an exit opening **157** and each of the cooling paths **154** are inclined with respect to the backstop **121** and the annular peripheral wall such that openings **155** further from the bottom of the annular peripheral wall **115** have a corresponding exit opening **157** further from the trailing edge **125** of the opening **116** at the bottom of the annular peripheral wall **115**. In other words, the higher inclination cooling paths **154** purge cooling air further aft of the bottom of the opening **116** as compared to cooling paths **154** at lower heights in the backstop **121** and/or annular peripheral wall **115**. In other words, the corresponding inlet openings **155** located closest to the hot side **112** of the liner panels **72**, **74** communicate with exit openings **157** furthest from the trailing edge **125** of a periphery of the bottom of the opening **116** defined by the annular peripheral wall **115**. In one non-limiting embodiment, the exit openings **157** may have shaped outlets to diffuse the cooling air at the hot side **112** of the liner panels **72**, **74** as well as prevent penetration of hot air back into openings **157** from the hot side **112** of the liner panels **72**, **74**. As such, openings **157** provide surface film cooling illustrated by area **146** in hard to cool regions behind the dilution hole **116** wake created by the cooling air provided in the direction of arrow **152**.

[0065] Referring now to FIGS. **8** and **9**, another alternative embodiment of the present disclosure is illustrated. This embodiment is similar to the embodiment illustrated in FIGS. **5** and **6** which includes protrusion **136** however, the internal plenum **130** is shorter in overall height than the internal plenum **130** of the embodiment of FIGS. **5** and **6**. In addition, the internal plenum **130** is angled with respect to the interior surface of the annular peripheral wall **115** defining the peripheral surface of the diffusion opening **116**. In this embodiment, an inlet or inlets **158** are located in the trailing edge **125** of the annular peripheral wall **115** as opposed to the backstop **121** and are located below the support shells **68**, **70** but above the liner panels **72**, **74**. As such, the inlet or inlets **158** are within the quench hole passage **116** defined by the annular peripheral wall **115**. As in the embodiment illustrated in FIGS. **5** and **6**, the inlet or inlets **158** are also in fluid communication with the combustion chamber **66** via a plurality of or an array of individual openings **140** located in a trailing edge portion **142** of protrusion **136**. In an alternative embodiment, the plurality of individual openings **140** may be replaced by a single elongated slot or opening **140**.

[0066] As such, the cooling air flow (illustrated by arrows **132**) will enter the interior plenum **130** and then exit the trailing edge portion **142** of the protrusion **136** via openings **140**. The exiting airflow is illustrated by arrows **144** This cooling air will provide surface film cooling to the hot side **112** of the liner panels **72**, **74**. This surface film cooling is illustrated by area **146**. This surface film cooling is downstream of the protrusion **136** and is particularly useful in cooling the hot side **112** of the liner panels **72**, **74** downstream of the protrusion **136**.

[0067] Referring now to FIGS. **10-11**B, another alternative embodiment of the present disclosure is illustrated. Here the backstop **121** and the annular peripheral wall **115** has an interior plenum **130**. The interior plenum **130** is in fluid communication with a cooling airflow (illustrated by arrows

132) via a plurality of openings **134**. The plurality of openings or at least a majority of the plurality of openings **134** are located in a surface of the backstop **121** above shells **68**, **70** so that they are positioned to capture cooling airflow flowing in the direction of arrows **132**, which may be located closer to a top **135** of the backstop **121**. This airflow above shells **68**, **70** and closer to a top **135** of the backstop **121** may be referred to as a velocity head of cooling airflow.

[0068] The annular peripheral wall **115** has a protrusion **136** that extends from a bottom portion **137** of the annular peripheral wall past the hot side **112** of the liner panels **72**, **74**. The distance the protrusion **136** extends past the hot side **112** of the liner panels **72**, **74** is illustrated by arrows **138**. The protrusion **136** like the backstop **121** partially circumscribes each or a subset of the dilution passages **116**. As illustrated, the protrusion **136** like the backstop **121** may have a curvature that matches a curvature of the dilution opening **116**. The interior plenum **130** extends through the backstop **121** and the annular peripheral wall **115** and is also in fluid communication with the combustion chamber **66** via a plurality of or an array of individual openings **140** located in a trailing edge portion **142** of protrusion **136**. In an alternative embodiment, the plurality of individual openings **140** may be replaced by a single elongated slot or opening **140**.

[0069] In addition, the interior plenum **130** is also in fluid communication with a plurality or array of openings **170** located in a surface the annular peripheral wall **115** arranged at a trailing edge portion **125** of the dilution passage **116**. The openings **170** are located in the opening **116** such that cooling air in the dilution opening **116** is fed into the interior plenum **130** (illustrated by arrows **172**) via cooling openings **170**.

[0070] As such, the cooling air flow (illustrated by arrows **132** and **172**) will enter the interior plenum **130** and then exit the trailing edge portion **142** of the protrusion **136** via openings **140**. The exiting airflow is illustrated by arrows **144** This cooling air will provide surface film cooling to the hot side **112** of the liner panels **72**, **74**. This surface film cooling is illustrated by area **146**. This surface film cooling is downstream of the protrusion **136** and is particularly useful in cooling the hot side **112** of the liner panels **72**, **74** downstream of the protrusion **136**.

[0071] In an alternative embodiment or configuration, a bottom surface **174** of the protrusion **136** facing the combustion chamber **66** is also provided with a plurality of openings **176** in fluid communication with the interior plenum **130** such that cooling air (illustrated by arrows **178**) may be provided to surface **174**. Openings **176** may comprises vanes, slots, or shaped holes. The openings **176** may be angled or normal to surface **174**.

[0072] As illustrated in FIG. **11**, the openings or single elongated slot or opening **140** are arranged in an arc **147** in order to provide the surface film cooling illustrated by area **146**. This area **146** is located upstream of the surface cooling provided by openings **104** and **108**, which is illustrated by arrows **148**, **150**. As illustrated in FIG. **10**, the cooling openings **108** are angled in a downstream orientation to provide cooling air in the direction of arrows **150**. This area **146** helps to reduce hot spots adjacent to dilution passages **116** which are hard to cool as they are directly behind the also permits utilization of the relatively limited cooling air elsewhere in the combustor allowing for the more efficient engine operation. There is also dilution air or cooling air that is jetted into dilution passage **116** that does not enter internal plenum **130**. This air is illustrated by arrow **152**.

[0073] In yet another alternative embodiment and referring now to FIGS. **10** and **11**A, the internal plenum **130** extends completely around opening **116** while being located in the annular peripheral wall **115**. This is in contrast to some of the preceding embodiments where the internal plenum **130** was only located inside some of the annular peripheral wall **115** that defines opening **116**. In addition, the entire inner surface of opening **116** including the leading edge portion **127** of the opening **116** is provided with openings **170** so that cooling air can enter plenum **130** about the entire periphery of the opening **116** in the direction of arrows **172**. In addition and similar to the previous embodiment, a bottom of the annular peripheral wall **115** as well as the bottom surface **174** of the protrusion **136** facing the combustion chamber **66** is also provided with a plurality of openings **176** in fluid communication with the interior plenum **130** such that cooling air (illustrated by arrows **178**) may be provided to surface **112**.

[0074] In yet another alternative embodiment and referring now to FIGS. **10** and **11**B, the protrusion **136** does not only extend partially from the bottom of the annular peripheral wall **115** but is located around the entire periphery of opening **116** or the bottom of the annular peripheral wall **115** essentially extending the entire annular peripheral wall **115** past surface **112** of the hot side **112** of the liner panels **72**, **74**. See the dashed lines in FIG. **10**.

[0075] In this embodiment and similar to the previous embodiment, the internal plenum **130** extends completely around opening **116**. In addition, the entire inner surface of opening **116** including the leading edge portion **127** of the opening **116** is provided with openings **170** so that cooling air can entire plenum **130** about the entire periphery of the opening **116** in the direction of arrows **172**. In this embodiment and since the protrusion **136** extends about the entire periphery of opening **116**, the bottom surface **174** about the entire periphery of the opening **116** facing the combustion chamber **66** is also provided with a plurality of openings **176** in fluid communication with the interior plenum **130** such that cooling air (illustrated by arrows **178**) may be provided to surface **174**.

[0076] As such, various embodiments of the present disclosure provide enhanced cooling about edges of the dilution openings **116**. In current designs, per-sector air flow pattern factors vary significantly, which requires a more conservative turbine cooling design. However, and by providing film cooling in accordance with the various embodiments of the present disclosure, the combustor exit temperature profile and uniformity can be controlled using the aforementioned design of air dilution passages **116**.

[0077] Since it is difficult to identify the sources of sector-to-sector variation and existing designs are potentially highly sensitive to manufacturing variations embodiments of the present disclosure are particularly useful in providing circumferentially uniform combustor exit temperature.

[0078] The various embodiments of the present disclosure may reduce variation in mixing between dilution jets, as the size and dynamics of the recirculation region could modulate the jet penetration and mixing.

[0079] In addition, the backstop **121** guides flow into and through the dilution passage **116**. The ensures that flow in and through each dilution passage **116** is the same, reducing passage-to-passage (and hence sector-to-sector) variation in mixing between dilution jets and combustor head end flow.

[0080] As such, the present disclosure provides an apparatus and method for providing uniformity to the combustor exit temperature profile, which in turn can increase turbine life and performance.

[0081] The use of the terms “a” and “an” and “the” and similar references in the context of description (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or specifically contradicted by context. The modifier “about” used in connection with a quantity is inclusive of the stated value and has the meaning dictated by the context (e.g., it includes the degree of error associated with measurement of the particular quantity). All ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. It should be appreciated that relative positional terms such as “forward,” “aft,” “upper,” “lower,” “above,” “below,” and the like are with reference to the normal operational attitude of the vehicle and should not be considered otherwise limiting.

[0082] Although the different non-limiting embodiments have specific illustrated components, the embodiments of this invention are not limited to those particular combinations. It is possible to use some of the components or features from any of the non-limiting embodiments in combination with features or components from any of the other non-limiting embodiments.

[0083] It should be appreciated that like reference numerals identify corresponding or similar elements throughout the several drawings. It should also be appreciated that although a particular component arrangement is disclosed in the illustrated embodiment, other arrangements will benefit

therefrom.
[0084] Although particular step sequences are shown, described, and claimed, it should be understood that steps may be performed in any order, separated or combined unless otherwise indicated and will still benefit from the present disclosure.
[0085] The foregoing description is exemplary rather than defined by the features within. Various non-limiting embodiments are disclosed herein, however, one of ordinary skill in the art would recognize that various modifications and variations in light of the above teachings will fall within the scope of the appended claims. It is therefore to be appreciated that within the scope of the appended claims, the disclosure may be practiced other than as specifically described. For that reason, the appended claims should be studied to determine true scope and content.

Claims

1. A wall assembly for use in a combustor of a gas turbine engine, the wall assembly comprising: a support shell; a liner panel; and an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage; a backstop extending from the top portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.
2. The wall assembly as in claim 1, further comprising an internal plenum located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum.
3. The wall assembly as in claim 2, further comprising a plurality of openings located in the surface of the backstop above the support shell, the plurality of openings being in fluid communication with the internal plenum.
4. The wall assembly as in claim 2, further comprising a protrusion that extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion.
5. The wall assembly as in claim 4, wherein the at least one opening is a plurality of individual openings or a single elongated slot.
6. The wall assembly of claim 5, wherein the protrusion has a curvature that matches a curvature of the dilution passage.
7. The wall assembly of claim 6, wherein the plurality of individual openings or the single elongated slot are arranged in an arc.
8. The wall assembly of claim 1, wherein the backstop is arranged to be at least partially located at a trailing edge portion of the dilution passage.
9. The wall assembly as in claim 1, further comprising a plurality of openings located in the surface of the backstop, a surface of the annular peripheral wall defining the dilution passage and a bottom portion of the annular peripheral wall, the plurality of opening being in fluid communication with the cooling airflow paths and the cooling airflow paths are angled with respect to the surface of the backstop and the bottom portion of the annular peripheral wall.
10. The wall assembly as in claim 9, wherein some of the plurality of openings located in the bottom portion the annular peripheral wall are further from a trailing edge of the dilution passage than other ones of the plurality of openings located in the bottom portion of the annular peripheral wall, the some of the plurality of openings located in the bottom portion of the annular peripheral

wall that are further from the trailing edge of the dilution passage correspond to some of the plurality of openings located in the surface of the backstop or the surface of the annular peripheral wall that are further from the bottom portion of the annular peripheral wall than others of the plurality of openings located in the surface of the backstop or the surface of the annular peripheral wall.

11. The wall assembly as in claim 1, further comprising: an internal plenum located in the annular peripheral wall, the cooling airflow paths extending through the internal plenum and at least one opening located in a surface of the annular peripheral wall defining the dilution passage, the at least one opening being in fluid communication with the internal plenum; a protrusion that extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion.

12. The wall assembly as in claim 11, wherein the at least one opening located in the trailing edge portion of the protrusion is a plurality of individual openings or a single elongated slot.

13. The wall assembly of claim 12, wherein the protrusion has a curvature that matches a curvature of the dilution passage.

14. The wall assembly of claim 13, wherein the plurality of individual openings or the single elongated slot are arranged in an arc.

15. The wall assembly as in claim 1, further comprising: an internal plenum located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum; and a plurality of openings located in the surface of the backstop above the support shell, a surface of the annular peripheral wall defining the dilution passage and a bottom portion of the annular peripheral wall, the plurality of openings being in fluid communication with the internal plenum.

16. The wall assembly as in claim 1, further comprising: an internal plenum located in the backstop and the annular peripheral wall, the cooling airflow paths extending through the internal plenum; a protrusion that extends from a bottom portion of the annular peripheral wall past the liner panel, the protrusion forms another portion a periphery of the dilution passage, the internal plenum extending into the protrusion and is in fluid communication with at least one opening located in a trailing edge portion of the protrusion; a plurality of openings located in the surface of the backstop above the support shell, and a surface of the annular peripheral wall defining the dilution passage, the plurality of openings and the at least one opening located in a trailing edge portion of the protrusion being in fluid communication with the internal plenum.

17. The wall assembly as in claim 16, wherein the protrusion completely encircles the dilution passage and a bottom portion of the protrusion has openings in fluid communication with the internal plenum.

18. The wall assembly as in claim 16, wherein the internal plenum extends completely around an interior of the annular peripheral wall.

19. A gas turbine engine, comprising: a compressor section; a turbine section; and a combustor section, the combustor section including a wall assembly for use in a combustor of the combustor section, the wall assembly comprising: a support shell; a liner panel; and an annular peripheral wall extending from the liner panel, the annular peripheral wall defining a dilution passage and the annular peripheral wall extends through an opening in the support shell when the support shell and the liner panel are secured to each other, the annular peripheral wall having a top portion that defines a portion of a first periphery of the dilution passage and a bottom portion that defines a portion of a second periphery of the dilution passage; a backstop extending from the top portion of the annular peripheral wall, the backstop defining another portion of a periphery of the dilution passage and the backstop extending through and above the opening in the support shell when the liner panel is secured to the support shell; and cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.

20. A method of guiding airflow into a combustor of a gas turbine engine, the method comprising: locating a portion of an annular peripheral wall in an opening of a support shell of the combustor, the annular peripheral wall extending from a liner panel secured to the support shell, the annular peripheral wall defining a dilution passage; and directing airflow into the dilution passage via a backstop extending from a top portion of the annular peripheral wall, the top portion defining a portion of a periphery of the dilution passage; and directing airflow into cooling airflow paths extending from a surface of the backstop through the backstop and the annular peripheral wall to provide cooling air to a surface of the liner panel.